

ARTIVA

Product Requirements Document

Version 1.7 — Full Depth Restored (Escrow + In-App Messaging Edition)
Group 5 | Cohort 8 | The Local Impact Challenge
Product Manager: Gabriel Ohigbamhudu Sunday
Date: August 2026 | Status: Draft — Open for Team Review

Document Control

Version	Date	Author	Status	Summary of Changes
1.0	Aug 2026	Gabriel O. Sunday (PM)	Superseded	First draft — business narrative, Abuja beachhead, escrow-lite model
1.1	Aug 2026	Gabriel O. Sunday + Team	Superseded	Merged technical scaffolding: API contracts, Firestore schema, screen inventory, track mappings, algorithms, demo script
1.2	Aug 2026	Gabriel O. Sunday + Team	Superseded	Five targeted fixes: no-response policy, subscription-model clarification, no-match risk mitigation, GPS scope wording, and numeric NFR targets
1.3	Aug 2026	Gabriel O. Sunday (PM)	Superseded	Brand rebrand to Artiva; logo and colour palette locked (Sage Teal #68A9AC, Gold #C9B121, Deep Navy #162D4F)
1.4	Aug 2026	Gabriel O. Sunday (PM)	Superseded	Full Artiva rebrand completed. WhatsApp contact removed. All negotiations and transactions are moved inside the app. In-app messaging was added to the MVP core loop.
1.5	Aug 2026	Gabriel O. Sunday (PM)	Superseded	Artisan signup flow finalised: 3-step onboarding (phone OTP + profile) → (trade + services) → (NIN + ID document + 3-5 work photos), per UI/UX hi-fi wireframes
1.6	Aug 2026	Gabriel O. Sunday (PM)	Superseded	Full merge pass: v1.5 decisions carried forward with restored v1.3 structural depth (escrow language, track mapping, API spec, constraints, GTM)
1.7	Aug 2026	Gabriel O. Sunday (PM)	Current	Full depth restored from v1.3 across every section — rich acceptance-criteria tables, Firestore data schema, priority-score algorithm, NFR targets, risk register, and open-questions log — while keeping all v1.4–v1.6 product decisions (escrow-lite, in-app messaging, NIN-based 3-step artisan signup, no WhatsApp hand-off).

Reviewer Sign-Off

Reviewer	Track	Review Due
Gabriel O. Sunday	UI/UX Design	Week 2, Day 1
Victor David	Backend Development	Week 2, Day 1
Victor David	Frontend Development	Week 2, Day 1

Ahmed Yetunde	Data Science	Week 2, Day 1
Reign- EL	Cybersecurity	Week 2, Day 1
Nnamdi	AI/ML	Week 2, Day 1
Ebele	Virtual Assistant	Week 2, Day 1
Ololade Ruth	Data Analysis	Week 2, Day 1
Ebele	Digital Marketing	Week 2, Day 1
Gabriel O. Sunday	Brand Design	Week 2, Day 1

1. Executive Summary

Artiva is a mobile-responsive web platform that connects homeowners and small contractors in Abuja with verified, available artisans for urgent home repairs. It addresses a two-sided market failure: clients cannot find a reliable artisan fast enough during an emergency, and skilled artisans cannot get consistent, well-paying leads.

The platform's core value proposition is speed and protection: a verified artisan is matched in under 30 minutes, with the client's payment held in escrow until the job is confirmed complete. It is not a directory — it is a trust-and-urgency marketplace. All negotiation, messaging, and transactions happen inside the app; there is no external WhatsApp or phone hand-off.

This PRD covers the MVP scope only: the minimum set of features required to prove the core loop end-to-end, generate real revenue, and demonstrate Product-Market Fit within a single beachhead estate — Life Camp (NTA Estate), Abuja.

Metric	Target
Q1 Revenue (conservative)	₦230,000
Q1 Revenue (optimistic)	₦362,500
Revenue target (hackathon brief)	₦200,000
Beachhead launch site	Life Camp (NTA Estate), Abuja
Phase 2 expansion	Karu, Mararaba (both NTA Properties estates)
Seed artisans at launch	30 across 6 trades
Client match fee	₦500 per match
Commission on jobs	10% of job value, held in escrow at booking
MVP timeline	4 weeks
Primary research	Artisans (n=47), Customers (n=105)

2. Problem Statement

2.1 The Problem

Nigeria's informal skilled labour market is fragmented, unregulated, and high-risk for both sides of the transaction. When a homeowner needs an urgent repair, the existing alternatives fail consistently in the one moment clients are most willing to pay for certainty.

The problem manifests as three connected pain points:

Pain Point	Client Impact	Artisan Impact	Evidence
Trust & Attendance deficit	Artisans' ghost jobs after contact; no-shows waste time and money	Reliable artisans are fully booked; new clients cannot find them	80.6% of Nigerian construction firms report a skilled labour shortage (2025 industry survey)
Communication black hole	Artisans go silent mid-project with no structured status update	Artisans lose repeat clients due to a poor communication infrastructure	Primary survey: The majority of customer respondents (n=105) reported ghosting incidents in the last 6 months
Financial vulnerability at the deposit stage	Upfront deposits are lost when artisans don't show or don't finish	Payment disputes damage an artisan's reputation and income reliability	23.1% of artisans (n=47) cite irregular income as their #1 barrier

2.2 The Insight

The real competitor is not another app. It is the estate WhatsApp group and the usual guy's phone number. Those alternatives work adequately when there is time to vet someone. They fail in the urgent moment — exactly when clients are most willing to pay for certainty.

Speed without trust is what WhatsApp groups already offer. Trust without speed is what existing artisan directories already own. The combination — verified and available within 30 minutes, with all communication and money protected inside the platform — is the gap nobody currently owns.

2.3 Gabriel's Domain Credibility

Gabriel Ohigbamhudu Sunday has run Oasis Interbiz Ltd, a construction and installation contracting business in Abuja, for 25 years. He has hired, fired, and managed artisans across every trade. He has the relationships to seed the first 15–20 artisans before any digital outreach begins, and he participated in the original construction of Life Camp — the beachhead estate.

3. Goals & Success Metrics

3.1 Product Goals

- Prove the core loop end-to-end, entirely inside the app: a real client posts a real job, matches a real verified artisan, pays a real ₦500 fee, negotiates and completes the job through in-app chat
- Generate a minimum of ₦200,000 in Q1 revenue from Life Camp alone
- Establish Artiva as the default option for urgent repairs within Life Camp before any competitor enters the estate
- Build a verified artisan pool of 30+ across 6 trades before client-side launch, so every search returns results

3.2 North Star Metric

Weekly Completed Jobs via Platform. This is the single metric that proves both sides of the marketplace are working: clients are finding value (jobs get done) and artisans are earning (jobs get completed).

3.3 KPIs

KPI	Definition	Week 2 Baseline	Month 1 Target	Q1 Target

Match conversion rate	Jobs posted where the client unlocked in-app chat with artisan ÷ total jobs posted	First real data point	40%+	50%+
Time to match	Time from job post to chat unlock	First real data point	<30 min median	<30 min median
Job completion rate	Matches where Mark Complete was triggered ÷ total matches	First real data point	60%+	70%+
Match fee revenue	Total ₦500 fees collected	₦0 (Week 1 build)	₦15,000	₦130,000
Commission revenue	Total 10% commission retained on completed jobs	₦0 (Week 1 build)	₦10,000	₦100,000
Seed artisan count	Verified artisans with a badge, approved and availability on	30 (Week 1 target)	30	45+
Artisan retention rate	Artisans active (≥1 lead received) in Month N ÷ total artisans	N/A	70%+	70%+
No-response rate	Matches where the artisan sent no in-app message within 4 hours ÷ total matches	First real data point	<20%	<10%

4. Users & Personas

4.1 Primary Users

User Type	Description	Primary Need	Willingness to Pay
Client	Resident of Life Camp or neighbouring NTA estate; owns or manages a property requiring maintenance	A verified artisan available now — not next week	Yes — ₦500 is a small price for certainty in an emergency
Artisan	Electrician, plumber, painter, bricklayer, welder, or aluminium fabricator operating in Abuja	Consistent, verified leads without standing at a junction all day	No upfront cost — earns through completed jobs on the platform (10% commission only)
Platform Admin	Oasis Interbiz team member manually reviewing artisan verification submissions	Fast, clear queue of pending verifications to process	Internal user — no payment

4.2 Persona 1 — Mrs Amaka (Client)

Attribute	Detail
Age & role	38, homeowner in Life Camp
Situation	The kitchen sink is leaking at 7 pm, before a family visit tomorrow morning
Current behaviour	Posts in the estate WhatsApp group; waits; gets one or two recommendations; calls them; one doesn't pick up, one wants to come 'next week'
Frustration	Has no way to verify who is trustworthy and no certainty that anyone will actually show up
What she needs	One tap to post; verified artisans available tonight; the ability to negotiate and pay entirely inside the app, with automatic refund protection if the artisan never responds
Willingness to pay	₦500 for guaranteed speed and recourse — yes, immediately

4.3 Persona 2 — Mr Emeka (Artisan)

Attribute	Detail
Age & trade	34, electrician, based near Life Camp
Situation	Spends 2–3 hours at the junction each morning, hoping for a client call
Current behaviour	Relies entirely on word-of-mouth referrals from past clients and neighbours
Frustration	Income is irregular; some weeks get 5 jobs, some weeks get none; no way to signal availability to new clients, and no way to negotiate without chasing clients on WhatsApp

What he needs	Leads sent directly to him; a verified badge that builds trust; an in-app way to negotiate and receive payment without chasing clients
Revenue model note	Mr Emeka pays nothing to join and nothing per month. He accepts 10% commission per completed job as the cost of a lead he would otherwise never receive. Earlier research explored a ₦3,000/month artisan subscription — that idea was tested, rejected, and is not part of this MVP.

4.4 Persona 3 — Gabriel (Admin Verifier)

Attribute	Detail
Role	Platform admin, also the Product Manager and Oasis Interbiz principal
Job-to-be-Done	Verify artisans quickly so the marketplace has supply credibility from day one
Context	He knows which artisans are reliable because he has hired and fired them for two decades. He can review a NIN, an ID document, and a photo of past work and determine competency — a skill no automated KYC system replicates at this stage.

5. Scope

5.1 In Scope — MVP

Core loop: Post urgent job → Match available verified artisan → Pay ₦500 match fee + full job value (held in escrow) → Negotiate & coordinate inside the app → Mark Complete → Release payment minus 10% commission → Rate artisan

Feature Area	In MVP	Notes
Client job posting	Yes	Trade + location + urgency (Today / This Week / Flexible) + short description
Artisan matching	Yes	Rule-based heuristic: trade filter + availability filter + distance sort + earned-priority tiebreaker + concentration-fix rotation weight
Artisan verification	Yes — manual	3-step signup: (1) phone OTP + name/experience/location, (2) trade + services, (3) NIN + ID document + 3-5 work photos; manual admin badge approval; 24-hour SLA
Escrow — match fee	Yes	₦500 via Paystack; held until Mark Complete
Escrow — job value / commission	Yes	Client pays full job value at booking; entire amount held by the platform; 10% retained on Mark Complete, remainder released to artisan
In-app messaging / negotiation	Yes	Simple text chat thread between client and matched artisan. All negotiation, scheduling, and status updates happen here. No external WhatsApp hand-off, no phone number reveal.
No-response policy	Yes	Auto-refund to client if artisan does not send an in-app message within 4 hours of match; artisan profile flagged for admin review
Mark Complete + rating	Yes	Client confirms job done; 1–5 star rating; triggers fund release; duplicate rating rejected with 409
Artisan dashboard	Yes	Held vs. released funds; jobs completed; reputation score; availability toggle; open chats
Admin verification queue	Yes	Pending list; approve/reject; no-response flags visible; admin UID environment-variable controlled
Edge & loading states	Yes	Empty match with Notify Me; failed payment with retry; unverified waiting screen; duplicate rating protection; spinner on every API-dependent screen
NDPR compliance	Yes	Consent on signup; data handling policy; NIN and ID documents restricted to admin UID
Analytics events	Yes	job_posted, artisan_matched, payment_initiated, payment_success, message_sent, job_completed, rating_submitted

5.2 Out of Scope — MVP

Feature	Reason	Roadmap Phase
WhatsApp deep-link / external chat/phone number reveal	Explicitly removed in v1.4 — all negotiation and coordination now happens inside the app, where it can be protected and audited	Removed permanently
AI/ML smart ranking	Needs 100–150 completed jobs minimum; rule-based heuristic performs well at launch density	Q3 — Platform
AI/ML fraud detection model	No training data exists at launch; rule-based input validation is sufficient for MVP	Q3 — Platform
AI/ML price estimator	No rate card dataset exists at launch; job value is self-reported for MVP	Q3 — Platform
Automated payout rail	Manual release on Mark Complete is sufficient for MVP	Q2 — Scale
Formal dispute resolution	7-day workmanship window + manual team review for MVP; Google Form escalation only	Q2 — Scale
Push notification infrastructure	Intent logging only in Week 2; no push infrastructure in MVP	Q2 — Scale
Multi-city launch	Density beats coverage; Life Camp first, then Karu/Mararaba via the same NTA Properties relationship	Q2 — Scale
Artisan insurance/loans	Strong survey interest but outside the MVP value proposition	Q3 — Platform
Full Dialogflow chatbot, multilingual support	A full chatbot with many intents, Igbo/Pidgin support, and a dedicated messaging platform is a standalone product build — not needed for the judged core loop	Q2 — Scale
Referral programme	Earned priority already incentivises quality; formal referral mechanics are a growth lever	Q2 — Scale
Precise GPS radius matching	Estate-level (Life Camp) location matching is used instead — sufficient for a single-estate beachhead where all artisans and clients are in the same area	Not planned
Automated KYC / biometric verification	Manual review by Gabriel is faster and more reliable at launch-scale; automated KYC is post-MVP	Q2 — Scale
iOS native app	Mobile-responsive web app only; PWA for installability	Q3 — Platform
Formal OWASP penetration test report	A focused security checklist and pre-demo sign-off are appropriate for MVP scope; a formal pentest is a post-launch commitment	Q2
Full STRIDE threat model document	Lightweight threat checklist implemented in Week 1 covers the 4 highest-risk surfaces adequately for MVP	Q2

6. User Stories & Requirements

6.1 Client Stories

ID	User Story	Acceptance Criteria	Priority
C-001	As a client, I want to post an urgent job in under 60 seconds so I can find help quickly	Form has 4 fields max; submits in under 60 seconds on low-end Android; job document created in Firestore on submit	Must Have
C-002	As a client, I want to see only artisans who are available right now	Match list filters on available = true; unavailable artisans never appear	Must Have

C-003	As a client, I want to see a verified badge so I know the artisan has been checked	Badge appears only when verified = true; unverified artisans never appear in match results	Must Have
C-004	As a client, I want to pay ₦500 to unlock in-app chat and be protected if the job fails	Paystack checkout renders, payment held; chat unlocks only after confirmed webhook	Must Have
C-005	As a client, I want to negotiate and coordinate with the artisan inside the app	In-app chat thread opens after payment; messages are stored; both parties can send text	Must Have
C-006	As a client, I want to mark a job complete so the artisan gets paid and I can rate them	Mark Complete triggers fund release minus commission; rating prompt appears; duplicate rating rejected with 409	Must Have
C-007	As a client, I want to see a clear empty state when no artisans are available	The empty state screen renders with a friendly message and a Notify Me button when zero results are returned	Must Have
C-008	As a client, I want to see a failed payment screen with a retry option	Failed payment screen renders on Paystack failure; Try Again re-initialises the same payment without requiring a new job post	Should Have
C-009	As a client, I want to be automatically refunded if my matched artisan never contacts me	If the artisan sends no in-app message within 4 hours of match confirmation, the platform initiates a Paystack refund; the client is notified in-app; the artisan's profile is flagged for admin review	Must Have

6.2 Artisan Stories

ID	User Story	Acceptance Criteria	Priority
A-001	As an artisan, I want to sign up with my phone number so I don't need an email	OTP sent to phone; verified on entry; artisan_profile document created on completion. Google/Apple buttons are shown as an optional sign-in affordance; phone OTP remains the functional MVP path.	Must Have
A-002	As an artisan, I want to select my trade and the specific services I offer so clients find me for the right job	Trade selected from a fixed list; at least one specific service selected within that trade before proceeding	Must Have
A-003	As an artisan, I want to submit my NIN, a valid ID document, and work photos so I can get my Verified badge	NIN captured as text; ID upload accepts NIN Slip / Driver's Licence / Voter's Card / International Passport (JPEG/PNG/PDF, server-side MIME validation); 3–5 work photos required; submission enters admin queue automatically	Must Have
A-004	As an artisan, I want to toggle my availability so clients only see me when I'm ready	Availability toggle updates the available field in real time; artisan disappears from match results when toggled off	Must Have
A-005	As an artisan, I want to see my held and released funds, so I know exactly what I've earned	Dashboard shows total held, total released, and completed job count; figures update within seconds of a Mark Complete event	Must Have
A-006	As an artisan, I want to build a reputation score through completed jobs, so I earn priority over time	Reputation score = weighted average of ratings; displayed on dashboard; used as a tiebreaker in matching sort	Must Have
A-007	As an artisan, I want to receive and reply to client messages inside the app after a match	Chat thread opens automatically once payment is confirmed; the artisan can send and receive text messages tied to that job	Must Have
A-008	As an artisan, I want to know if a lead was refunded due to my non-response, so I can maintain my profile standing	If a refund is triggered due to non-response, the artisan receives an in-app notification; an admin flag is logged against their profile	Must Have

6.3 Admin Stories

ID	User Story	Acceptance Criteria	Priority
AD-001	As an admin, I want to see a queue of pending verifications so I can approve artisans quickly	Admin queue lists all artisans with verified = false; NIN, ID document, and work photos viewable on tap; Approve sets verified = true; only admin UID can access	Must Have
AD-002	As an admin, I want pending verifications to clear within 24 hours	Queue monitored daily; target: zero profiles unreviewed for more than 24 hours	Must Have
AD-003	As an admin, I want to view transaction logs and revenue metrics to prove business viability to judges	Admin dashboard shows: total match fees, total commission, jobs completed today, active artisans, conversion funnel	Must Have
AD-004	As an admin, I want to see artisan profiles flagged for non-response so I can take action	Non-response flags appear on the artisan profile in the admin queue; the admin can warn or remove the artisan	Must Have

7. Functional Requirements

7.1 Authentication & Artisan Onboarding

- Phone OTP via Firebase Auth — no email required
- OTP requests are rate-limited to 3 per phone number per hour; 24-hour lockout after 5 failed attempts
- Session token required on every authenticated request
- Admin access controlled by environment-variable admin UID — not a role field in the database
- Artisan 3-step signup: (1) phone OTP + name/experience/location, (2) trade + specific services, (3) NIN + ID document + 3-5 work photos + NDPR consent
- NIN is stored securely for verification only — never shown to clients. Accepted ID documents: NIN Slip, Driver's Licence, Voter's Card, International Passport
- Google/Apple continue buttons are shown on signup for UX polish; Firebase phone OTP remains the only functional auth path for MVP (flagged for team review — full OAuth is a post-MVP add if time allows)

7.2 Matching Engine

1. Hard filter: trade = job.trade AND available = true AND verified = true
2. Sort: distance ascending (closest first)
3. Tiebreaker: completed_jobs descending, then reputation_score descending (null = sorted last)
4. Zero-result handling: return an empty list with an empty-state trigger; never return unverified artisans
5. Concentration fix: rotation weight applied to prevent the same artisan from dominating consecutive matches in the same estate area

Note: The matching engine is rule-based by design. A learned ranking model is scoped for Q3 once 100–150 real completed jobs provide sufficient training signal.

Priority Score Algorithm (v1)

$priority_score = (avg_rating \times 0.40) + (min(completed_jobs, 50) / 50 \times 0.30) + (response_speed_score \times 0.20) + (verification_bonus \times 0.10)$

response_speed_score: 1.0 if avg response < 30 min; 0.5 if < 2 hours; 0.2 if < 24 hours; 0.0 otherwise

verification_bonus: 1.0 if verified = true; 0.3 if verified = pending

No-Match Risk Mitigation

Risk Scenario	Mitigation	Owner
A trade category has zero seed artisans in Life Camp	Virtual Assistant and Digital Marketing onboard artisans by trade category before client-side launch; Data Science's simulation identifies zero-result trades each week so outreach can target specifically	Virtual Assistant + Digital Marketing
All artisans in a trade are marked Busy simultaneously	Rotation weight and availability toggle remind artisans to update their status; admin monitors availability distribution weekly	Admin / Product Management
Client location doesn't match any artisan's service area	MVP uses estate-level (Life Camp) rather than precise GPS, so all artisans and clients in the beachhead share the same location value, reducing zero-result risk significantly compared to a city-wide model	Backend / Data Science
Minimum viable supply threshold not met	The client-side platform does not launch until at least 20 verified artisans across all 6 core trades are confirmed. If supply is below this threshold, launch is delayed by 48 hours rather than launching with predictably empty results.	Product Management

7.3 Escrow & Payment Flow

Artiva operates an escrow-lite model: the client pays the full job value plus the ₦500 match fee at booking, and the entire amount is held by the platform until the job is confirmed complete.

Step	Action	System Behaviour
1	Client taps Match Now — ₦500 + job value	Paystack payment initialised server-side; job_value locked in transaction document; no-response timer started (4-hour countdown)
2	Client completes Paystack checkout	Paystack fires a webhook to the platform on success
3	Platform receives webhook	Signature (HMAC) verified; transaction status updated to held; in-app chat thread opened between client and artisan
4a — Normal path	Client taps Mark Complete	Authorization checked; commission calculated server-side (locked_job_value × 0.10); artisan share released; transaction status updated to released; no-response timer cancelled
4b — No-response path	Artisan sends no in-app message within 4 hours of the match	Platform initiates automatic Paystack refund to client; transaction status updated to refunded; artisan profile receives a non-response flag; admin notified; client sees an in-app message explaining the refund
5	Rating submitted	1–5 star value stored; reputation_score recalculated; duplicate submission rejected with 409
6	Payment failure	Failed payment screen rendered; retry re-initialises the same payment; job remains open

Note: Definition of "contact": the first message sent by the artisan inside the Artiva chat thread. Opening the app alone does not count. The 4-hour window is deliberate: short enough to protect the client from being left waiting, long enough to account for artisans who are mid-job when a lead arrives. The window is enforced by a server-side timer (Firebase scheduled function), not dependent on client action.

7.4 In-App Messaging

- Simple real-time chat (Firebase Firestore or equivalent) between matched client and artisan only
- Chat becomes available only after successful match-fee and job-value payment
- Messages are text-only in MVP (no media/file sharing required for launch)
- Both parties can see message timestamps and read status (basic)
- Chat history is retained for the life of the job and visible to admin for dispute context
- No WhatsApp deep-link, no phone number reveal, no external hand-off

7.5 Data Requirements (Firestore Schema)

--	--	--

Collection	Key Fields	Access Rules
users	uid, phone (OTP-verified), role (client/artisan/admin), created_at	User reads/writes own document only
artisan_profiles	uid, trade, services[], nin, id_document_url, work_photos[], location, available, verified, completed_jobs, reputation_score, locked_job_value, no_response_flags (integer)	Artisan writes own profile; clients read verified profiles only (never nin or id_document_url); admin writes verified field and no_response_flags
jobs	job_id, client_uid, trade, location, urgency, status (open/matched/complete/refunded), created_at	Client writes own jobs; artisans read open jobs in their trade/location
matches	match_id, job_id, artisan_uid, client_uid, status, no_response_timer_expiry, created_at	Client and matched artisan read; platform writes on payment confirmation
messages	message_id, match_id, sender_uid, text, sent_at, read_at	Client and matched artisan on that match read/write; admin reads for dispute context
transactions	transaction_id, match_id, amount, locked_job_value, commission, status (held/released/refunded), paystack_reference, refund_reason, created_at	Client and artisan read own transactions; platform writes on webhook, Mark Complete, and no-response refund
ratings	rating_id, job_id, artisan_uid, client_uid, score, review, created_at	Client writes once per job; artisan reads own ratings
analytics_events	event_id, event_type, user_id, session_id, metadata, timestamp	Platform writes; Data Analysis reads

8. Track Feature Mapping

This section maps each learning track to their specific PRD features, deliverables, and acceptance criteria. Scope is set to what can genuinely be built and demoed in 4 weeks.

Track	MVP Deliverables	Acceptance Criteria
Product Management	Sprint planning; backlog grooming; daily standups; problem statement; judging criteria map; beachhead confirmation; demo script; pitch rehearsal coordination	Every card moves through backlog → in progress → done on schedule; no blocker open >24hrs without escalation
UI/UX Design	User research synthesis (8 interviews); user flows; low-fi wireframes (10 screens); hi-fi prototypes; design system; usability test reports (2 rounds)	All screens mobile-first (375px); <60 seconds to post job; Lighthouse accessibility >90
Brand Design	Logo (Artiva sage-teal circular badge + gold speed lines, tested at 48x48px); colour palette (Sage Teal #68A9AC, Gold #C9B121, Deep Navy #162D4F); marketing assets; pitch deck visual template; 12-slide final deck; brand guidelines	Final asset pack confirmed by Day 3; all 12 slides built with real screenshots and data by Week 3
Backend Development	Database schema; API spec; Firebase project; Phone OTP; Paystack sandbox + live integration; escrow logic; no-response timer (Firebase scheduled function); in-app chat APIs; admin queue; webhook handling; API documentation	All P0 endpoints live by Day 14; Paystack test transaction successful by Day 7; no-response timer tested by Day 21
Frontend Development	React + Tailwind + Vite scaffold; all 10 MVP screens; responsive pass (320→414px); Paystack checkout; in-app chat UI; edge states; loading states on every API-dependent screen	End-to-end loop working on low-end Android by Day 21; Lighthouse performance >85
Cybersecurity	Lightweight threat checklist; secure coding guidelines (1 page); hardened Firestore rules; input validation layer (Zod); Paystack webhook signature verification; NIN/ID access restriction; pre-demo security sign-off; NDPR compliance note	0 critical vulnerabilities in pre-demo sign-off; no secrets in git history; all 4 highest-risk surfaces confirmed secure by Day 21
Data Science	Data dictionary; matching heuristic v1 (plain-language spec + implementation); synthetic test dataset; validation script; concentration-fix validation; revenue model spreadsheet with matching-constraints and reputation-effect tabs	Matching algorithm deployed and validated against 5-location test by Day 14; accuracy report by Day 21; zero-result trades identified weekly

Data Analysis	Metrics framework; event taxonomy; Firebase Analytics dashboard; custom event instrumentation; live metrics dashboard showing revenue and conversion; Q1 projection model with downside case	Live dashboard receiving real event data from Week 2; all KPIs in Section 3.3 tracked with at least one real data point by Day 28
AI/ML	Research only in Week 1: feasibility note on improving the matching heuristic post-MVP. Week 2: monitor real job completion data. Deliverable: one-page feasibility note + Q3 roadmap slide text. No code shipped to production.	Feasibility note complete by Day 5; Q3 roadmap slide text confirmed with PM by Day 4; no AI/ML code in production during the 4-week sprint
Digital Marketing	Brand voice guide; 4-week content calendar; artisan outreach playbook; NTA Properties estate partnership pitch; beachhead partnership pitch; artisan onboarding campaign; resident-facing messaging (ready for NTAPIC approval)	30 artisans onboarded by Day 14; NTAPIC estate partnership engaged by Day 7; resident-side campaign live within 48hrs of NTAPIC approval
Virtual Assistant	FAQ document (7–10 questions); saved-reply script; artisan onboarding log; admin verification queue coordination; real job completion log	30 artisans confirmed onboarded and verified by Day 14; zero profiles unreviewed >24hrs in admin queue from Week 2 onward

9. API Specification

9.1 Authentication

Endpoint	Method	Description	Response
/api/auth/phone/send-otp	POST	Request OTP to phone number	{ success: boolean, message: string }
/api/auth/phone/verify-otp	POST	Verify OTP and create session	{ token: string, user: User }

9.2 Artisans

Endpoint	Method	Auth	Description
/api/artisans	POST	Required	Create artisan profile (3-step signup payload)
/api/artisans	GET	Optional	List artisans with filters (trade, location, available)
/api/artisans/:id	GET	Optional	Get artisan profile detail (excludes nin, id_document_url for non-admin)
/api/artisans/:id/availability	PATCH	Artisan only	Toggle availability — owner check required

9.3 Jobs & Matching

Endpoint	Method	Auth	Description
/api/jobs	POST	Client only	Create job posting
/api/jobs/:id	GET	Owner only	Get job details
/api/jobs/:id/matches	GET	Client only	Get matching artisans (heuristic applied)
/api/jobs/:id/complete	POST	Client only	Mark job complete + submit rating

9.4 Payments & Escrow

Endpoint	Method	Auth	Description
/api/payments/initialize	POST	Client only	Initialize Paystack payment; lock job_value; start 4-hour no-response timer

/api/payments/webhook	POST	Public (signed)	Paystack webhook callback; verifies HMAC signature before processing; opens chat thread
/api/payments/refund	POST	Platform only (internal)	Initiates refund on no-response trigger; updates transaction status to refunded; flags artisan profile. Not callable by clients or artisans.

9.5 Chat & Admin

Endpoint	Method	Auth	Description
/api/chat/:matchId/messages	GET / POST	Match participants only	List / send messages in the match thread
/api/admin/verification-queue	GET	Admin only	List pending artisan verifications
/api/admin/verify/:artisanId	POST	Admin only	Approve or reject artisan
/api/admin/analytics	GET	Admin only	Dashboard metrics: users, jobs, matches, revenue, no-response rate
/api/admin/flags	GET	Admin only	List artisan profiles with no-response flags

10. Technical Architecture

Layer	Technology	Rationale
Frontend	React 18 + Tailwind CSS + Vite	Mobile-first, fast builds, credible demo UI
Backend / API	Node.js + Express + Firebase Firestore	Auth/OTP out-of-box, real-time DB, generous free tier
Authentication	Firebase Auth (phone OTP)	SMS OTP for Nigerian numbers without an additional vendor
Payments / Escrow	Paystack (sandbox → live)	Nigerian-native payment rails, sandbox available, judge-friendly; funds held platform-side until Mark Complete
Chat	Firestore real-time listeners (simple text threads)	No separate messaging vendor needed; ties naturally to existing Firestore auth and security rules
Scheduled functions	Firebase Cloud Functions (scheduled)	No-response timer; runs server-side, not dependent on client action
File Storage	Firebase Storage	Integrated with Firestore auth; server-side MIME validation before write
Hosting	Vercel (Frontend) + Firebase Functions (API)	Free tier, auto-deploy on git push, Nigeria-accessible CDN
Analytics	Firebase Analytics + custom events	Match conversion, time-to-match, job completion rate tracked from Week 2
Design	Figma (prototypes) + FigJam (workshops)	Collaborative, export-ready
Project Management	Notion / Trello + daily standups	Sprint boards; retrospectives

11. Non-Functional Requirements

11.1 Performance

Requirement	Target	Measurement Method
Match list load time	<3 seconds on 3G mobile connection (approx. 1Mbps)	Lighthouse + manual test on real Android device over 3G/4G
API response time (p95)	<500ms for all P0 endpoints	Firebase Functions logs; Backend monitoring

Payment flow (tap to Paystack redirect)	<10 seconds	Frontend event timing from <code>match_now_tapped</code> to <code>paystack_opened</code>
Image loading	Progressive JPEG; lazy load below fold; max 200KB per photo after compression	Lighthouse + Firebase Storage upload rule
Time from job post to first match result	<5 seconds	Backend log: <code>job_posted</code> → <code>matches_returned</code>
Lighthouse performance score	>85 on mobile	Lighthouse CI run in Week 3 and Week 4

11.2 Scalability

Requirement	Target	Notes
Concurrent users (MVP)	100 concurrent users without degradation	Firestore + Vercel free tier handles this without optimization at beachhead scale
Artisan profiles (MVP)	Up to 200 profiles	Firestore free tier: 1GB storage, well within range
Simultaneous match flows	Up to 20 concurrent matching requests	Firebase Functions auto-scales; no custom queue needed for MVP
Payment concurrency	Up to 10 simultaneous Paystack transactions	Paystack handles concurrency; idempotency key prevents double-charge
File uploads	Up to 10 concurrent uploads (artisan verification documents)	Firebase Storage handles concurrency; MIME validation before write

11.3 Reliability

Requirement	Target	Fallback
Uptime during demo period (Week 4)	99% — less than 7 minutes downtime across the demo window	If live app fails: pre-recorded screen capture of complete demo path ready as backup
Paystack availability	Not within team's control — Paystack SLA is >99.9%	Manual bank transfer fallback documented and tested for first 10 match fees
No-response timer reliability	Firebase scheduled function fires within 5 minutes of 4-hour expiry	Admin manually triggers refund if function fails; monitored via Firebase console
API cold start time	<2 seconds for Firebase Functions cold start	Functions kept warm during demo window via scheduled ping
Offline behaviour	Static assets cached via service worker; dynamic content requires connection; loading states on all API-dependent screens	Error states with retry on all failure paths; never a blank screen
Data backup	Firestore automatic daily backups enabled before Week 3 live key activation	Point-in-time recovery available for up to 7 days

11.4 Security

Category	Requirement
Payments	Commission calculated server-side from <code>locked_job_value</code> ; client-side amount cannot affect commission
Authentication	Every endpoint checks requesting user's UID against resource owner before responding (IDOR prevention)
File uploads	Server-side MIME type validation; extensions not trusted
Webhooks	Paystack HMAC signature verified on every incoming request before processing
Data privacy	NIN and ID documents accessible only to admin UID; never in client-facing API responses
Secrets management	All keys in environment variables; <code>.env</code> in <code>.gitignore</code> from Day 1; no secrets in git history
OTP abuse	Rate limited: 3 requests per hour per phone; 24-hour lockout after 5 failed attempts

NDPR compliance	User consent on signup; data handling policy published; ID documents deleted after verification on request
-----------------	--

11.5 Accessibility & Responsiveness

Requirement	Standard / Target
Contrast ratio	WCAG AA — 4.5:1 minimum for all text; especially critical for outdoor Nigerian daylight use
Touch targets	Minimum 44×44px for all interactive elements
Font sizes	Minimum 16px for inputs (prevents iOS zoom)
Screen reader support	Alt text on all images; semantic HTML throughout
Responsive breakpoints tested	320px (low-end Android), 375px (design baseline), 414px (large Android), 768px (tablet), 1024px (desktop admin panel)
Test device	Real Tecno or Infinix phone by Week 2 in addition to browser simulation

12. Revenue Model

12.1 Revenue Streams

Stream	Amount	When Collected	How Enforced
Client match fee	₦500 per match	At booking, before chat is unlocked	Paystack payment required; chat unlock gated on successful webhook
Commission on job value	10% of locked job value	Retained automatically on Mark Complete	Full job value held in escrow from booking; artisan share released minus commission; nothing to invoice or chase

12.2 Q1 Projections

Scenario	Q1 Total	Multiple of Target
Conservative	₦230,000	1.15×
Optimistic	₦362,500	1.81×
Downside (85% of conservative volumes)	≈₦195,500	0.98×

Note: Downside case at 0.98× target is within one additional week of Month 3 push. Month 3 network effects have the most room to close the gap.

12.3 Why Commission Cannot Be Skipped

The client pays the full job value plus the match fee at booking. The platform holds the entire amount in escrow. On Mark Complete, the platform releases the artisan's share (job value minus 10%) and retains the commission. The artisan cannot receive their share without the job being marked complete, and the commission is never invoiced separately — it is retained automatically from funds already held.

13. Go-to-Market Strategy

13.1 Beachhead — Life Camp (NTA Estate), Abuja

Life Camp is not a randomly selected launch site. NTA Properties and Investment Company (NTAPIC) owns the estate, and Oasis Interbiz participated in its original construction. Gabriel visited NTAPIC headquarters and spoke directly with Mr. Aminu (internal champion). A formal letter was submitted to the Managing Director on Day 3 of the sprint. This is a developer-level relationship, not a cold pitch.

13.2 Why Beachhead Over City-Wide Launch

Launching across Abuja with 30 artisans produces approximately one artisan per square kilometre — too thin to guarantee results on any search. Concentrating those same artisans in one estate means every search returns results, every completed job becomes word-of-mouth within the same resident group, and the platform's density advantage compounds before any competitor can copy it.

13.3 Phase Roadmap

Phase	Target	Timeline	Revenue Model
MVP (Q1)	Life Camp (NTA Estate), Abuja — 200+ households, 30 verified artisans	Weeks 1–4	₦500 match fee + 10% commission
Scale (Q2)	Karu + Mararaba (NTA Properties); automated payout rail; referral mechanic	Months 4–6	Same + referral mechanic
Platform (Q3)	City-wide Abuja; artisan tools; ML ranking; full chatbot	Months 7–9	Match fee + commission + subscription options

14. Constraints & Assumptions

14.1 Constraints

Constraint	Impact	Mitigation
4-week build timeline	MVP scope must be genuinely buildable; no feature creep	Scope line enforced at PM level; ponytail principle applied to every feature discussion
Student team — no dedicated DevOps	No auto-scaling, no CI/CD pipeline beyond git push	Vercel auto-deploy sufficient for MVP; manual monitoring during demo window
Manual artisan verification	Admin bandwidth limits verification throughput	24-hour SLA per submission; admin queue built in Week 1
Paystack sandbox in Weeks 1–2	Cannot process real money until live keys enabled	Live keys enabled before Week 3; manual bank transfer fallback for first 10 match fees
NTAPIC partnership pending	Resident-side launch dependent on MD approval	Mr. Aminu as internal champion; formal letter submitted Day 3; Gabriel relationship active
Estate-level matching with thin supply	Zero-result searches possible if artisan pool is thin in specific trades	See Section 7.2 No-Match Risk Mitigation; minimum 20 artisans across all 6 trades before client launch

14.2 Assumptions

- Life Camp has sufficient household density (200+) to generate 30 match-fee transactions in Month 1
- Gabriel's contractor network can supply at least 15 seed artisans willing to onboard for free in Week 1
- NTAPIC approval for resident outreach arrives before Week 2's client-side launch
- The average job value for urgent home repairs in Life Camp is approximately ₦20,000 — the basis for commission projections
- Artisans are willing to accept 10% commission per job in exchange for leads they would not otherwise receive
- Paystack live keys can be enabled within the 4-week sprint without additional business registration
- The 4-hour no-response window is acceptable to both clients (willing to wait up to 4 hours for a verified artisan rather than immediately refund) and artisans (4 hours is sufficient to respond between jobs) — to be validated with real users in Week 2

15. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation	Owner
	Medium	High		Digital Marketing

NTAPIC MD does not respond in time for Week 2 client launch			Mr. Aminu as internal champion; Gabriel personal relationship; fallback: direct resident outreach via Gabriel's Life Camp contacts	
Artisans take jobs off-platform after first contact	Medium	High	Commission held in escrow at booking; earned priority makes on-platform history valuable	Product Management
Seed artisan count below 30 at client launch	Medium	High	Minimum 20 across all 6 trades before launch; delay 48hrs rather than launch with empty results	Virtual Assistant + Gabriel
Frequent zero-result searches due to thin supply	Medium	Medium	No-match risk mitigation (Section 7.2); Data Science identifies zero-result trades weekly; estate-level matching reduces risk vs. city-level	Data Science + Digital Marketing
Client pays but artisan never responds in-app	Medium	High	4-hour auto-refund policy (Section 7.3); artisan profile flagged for admin review on non-response; client sees in-app explanation	Backend + Product Management
Paystack integration fails during live demo	Low	Critical	Manual bank transfer fallback documented and tested; Paystack sandbox rehearsed 10+ times before demo day	Backend Development
Reputation score concentration	Low — mitigated	Medium	Rotation weight validated with 5-location test in Week 1	Data Science
AI/ML scope creep into MVP build	Low — mitigated	High	AI/ML track explicitly scoped as research only in Week 1; monitoring only in Week 2; no code in production	AI/ML + Product Management
Scope creep on in-app chat feature	Low — mitigated	Medium	Keep messaging text-only and match-gated; no media in MVP	Frontend + Product Management
Revenue downside scenario (₦195,500 vs ₦200,000 target)	Low	Low	85% downside is within one additional week of Month 3 push	Data Analysis

16. Open Questions for Team Review

#	Question	Status	Decision / Owner	Due
Q01	What is the exact definition of 'contact' for the no-response timer — must the artisan send an in-app message, or does opening the chat thread count?	Closed	Resolved in v1.4: only a sent message counts as contact. Opening the app alone does not.	Closed
Q02	Should the 4-hour no-response window be configurable (e.g., shorter for 'Today' urgency, longer for 'This Week' urgency)?	Open	PM + Data Science	Week 2, Day 2
Q03	What happens if a client marks Complete but the artisan disputes the job quality? Who reviews, and what is the outcome — refund, redo, or mediation?	Open	PM	Week 2, Day 2
Q04	Should the empty-state Notify Me intent also alert admin when a trade is consistently returning zero results, so onboarding can target that trade proactively?	Open	Backend + Virtual Assistant	Week 2, Day 2
Q05	At what artisan count per trade does the rotation weight need to be adjusted — is 5 artisans per trade sufficient for fair rotation, or is a higher minimum needed?	Open	Data Science	Week 2, Day 3
Q06	Should the NDPR consent statement on the artisan signup screen be a checkbox or a passive notice? Legal implications differ.	Open	PM + Cybersecurity	Week 2, Day 1
Q07	Team Member 2 and Team Member 3 are still placeholders on the pitch deck. Names and roles must be confirmed before any external pitch rehearsal.	Open	PM	Week 2, Day 1
Q08	Beachhead estate: Life Camp or an alternative?			Closed

		Closed — 2026-08-03	Life Camp (NTA Estate), Abuja confirmed. Gabriel visited NTAPIC headquarters Day 3; formal letter submitted to MD.	
Q09	Paystack live account approval timeline?	In progress	Sandbox ready Day 1. Live account application submitted Day 3. Expected Week 3.	Week 2, Day 7
Q10	Should artisan profiles with 2+ no-response flags be automatically suspended pending admin review, or should this remain a fully manual decision?	Open	PM + Backend	Week 2, Day 3
Q11	Google/Apple sign-in buttons appear on the finalized signup wireframes but are not part of the functional MVP auth path — build full OAuth, or keep as a visual affordance only?	Open	PM + Backend	Week 2, Day 1

17. Brand & Design Tokens

Name: Artiva | Tagline: Verified. Fast. Protected.

Primary (Sage Teal): #68A9AC

Accent (Gold): #C9B121

Text / Navy: #162D4F

Full Tailwind config + CSS custom properties already delivered to Frontend.

18. Glossary

Term	Definition
Beachhead	A single, highly concentrated launch market chosen to prove the model before expanding
Contact	The first message sent by the artisan inside the Artiva chat thread. Opening the app alone does not count.
Earned priority	Artisan ranking improvement earned through completed jobs and positive ratings — no subscription fee
Escrow-lite	The MVP payment mechanism: client's full payment held by the platform, released to artisan on Mark Complete, minus commission
IDOR	Insecure Direct Object Reference — a security vulnerability where a user can access another user's resources by manipulating IDs
MIME type	The actual file format as read from the file's binary signature, not the file extension — used for server-side file upload validation
NDPR	Nigeria Data Protection Regulation — the Nigerian equivalent of GDPR governing collection and use of personal data
NIN	National Identification Number — captured during artisan verification, stored securely, and never shown to clients
No-response timer	A 4-hour server-side countdown started at match confirmation; triggers an automatic refund to the client if the artisan sends no in-app message before expiry
Rotation weight	A matching heuristic adjustment that prevents the same artisan appearing first for every job posted from the same estate area
Verified badge	A visual trust signal on an artisan's profile, issued manually by the platform admin after OTP, NIN, ID, and work-photo review

19. Appendix A: Screen Inventory (10 MVP Screens)

--	--	--	--	--

#	Screen	User	Purpose	Key Elements
1	Splash / Onboarding	All	First impression, value prop	Logo, tagline 'Verified. Fast. Protected.', Get Started, I am an artisan
2	Client: Post Job	Client	Create urgent job posting	Trade dropdown, description, location (default: Life Camp), urgency chips (Today/This Week/Flexible)
3	Client: Browse Artisans	Client	View matched artisans	Cards with verified badge, rating, response estimate, Match CTA
4	Client: Artisan Profile	Client	Evaluate artisan before paying	Photos, stats, reputation score, Match Now — ₦500 CTA (no phone number shown)
5	Client: Paystack Checkout	Client	Complete escrow payment	Paystack inline checkout, ₦500 + job value display, 'held until job complete' protection note
6	In-App Chat	Both	Negotiate and coordinate	Message thread, timestamps, read status, Mark complete entry point
← Artiva_PRD_v1_7				Complete button, 1–5 star rating, optional review, confirm release
8	Artisan Signup (3-step)	Artisan	Create verified profile	Step 1: phone OTP + name/experience/location (Google/Apple affordance). Step 2: trade + multi-select services. Step 3: NIN + ID upload + 3-5 work photos + NDPR consent → admin queue
9	Artisan Dashboard	Artisan	View performance and earnings	Held vs. released funds, jobs completed, reputation score, availability toggle, open chats
10	Admin: Verification Queue	Admin	Review and approve artisans	Pending profiles list, NIN/ID viewer, work photo gallery, Approve/Reject buttons, rejection reason text, no-response flags visible

20. Appendix B: Demo Script (10-Minute Pitch)

Time	Action	Speaker	Screen / Slide
0:00	Hook: Mrs. Amaka's kitchen sink is leaking at 7pm...	Gabriel	Slide 2
1:00	Problem data (80.6%, 23.1%, n=47, n=105)	PM	Slide 3
2:00	Solution: end-to-end loop, all inside the app	PM	Slide 4
3:00	LIVE DEMO: Client posts urgent job	Frontend	Screen 2: Post Job
3:30	LIVE DEMO: Browse available verified artisans	Frontend	Screen 3: Browse
4:00	LIVE DEMO: Tap Match Now — ₦500	Frontend	Screen 4: Profile
4:30	LIVE DEMO: Paystack payment succeeds; chat unlocked	Frontend	Screen 5: Checkout
5:00	LIVE DEMO: In-app chat opens between client and artisan	Frontend	Screen 6: In-App Chat
5:30	LIVE DEMO: Artisan dashboard (reputation building)	Frontend	Screen 9: Dashboard
6:00	Revenue model: match fee + commission held in escrow at booking	PM / Data Analysis	Slide 7
7:00	Q1 projections: ₦230K conservative	Data Analysis	Slide 8
7:30	Beachhead: Life Camp, NTAPIC, 30 artisans, Phase 2 Karu + Mararaba	Digital Marketing	Slide 9
8:00	Competitive differentiation: we own the urgent moment, protected inside the app	PM	Slide 10
8:30	Team: Gabriel's 25 years in Abuja on the very estates we're launching in	Gabriel	Slide 11
9:00	Evolution roadmap (MVP → Scale → Platform)	PM	Slide 12
9:30	The ask: commit to this as our capstone direction	Gabriel	Close slide

21. Single Source of Truth

This document (Artiva PRD v1.7) is the single source of truth for the 4-week sprint. All tracks must reference it for acceptance criteria. Any change requires PM approval and notification to all track owners.